Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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Abstract

A decoder fit to one animal's cortex loses accuracy on that animal over days, and the usual remedy is to recalibrate it on fresh data from the same subject. We report a different kind of robustness: a decoder built from *other* individuals ages more slowly than the subject's own. On the open BraiDyn-BC cohort (25 mice, 44 Allen-atlas cortical parcels, a $\sim$15-day operant protocol) we decode four task events and measure, per mouse, the AUC lost over two weeks by its own early-week decoder versus one pooled over the other 24 mice. The personal decoder ages more by $+0.017$ AUC ($p = 5 \times 10^{-5}$, 73 of 100 mouse–target pairs). The size of the effect varies across the four targets, but they are chained within a trial – every reward onset has a lick within $0.12\,\mathrm{s}$ – and differ tenfold in sample size, so we report that variation without attributing it to the behaviors. At matched training-event count a decoder from a single other mouse still ages less, so this is not a data-volume effect but a consequence of not overfitting the subject's drift-prone idiosyncrasies; it survives a 1-D CNN and a GRU as well. One caveat is load-bearing: trials within a session are not independent, so splitting at the trial level inflates the personal decoder alone, by up to 0.013 AUC – enough to manufacture the entire lever-pull effect. All estimates here hold out whole sessions. Code and a streamed-feature pipeline are released.

1 Introduction

Representational drift, the reorganization of an animal's neural code over days with fixed behavior, is now documented across sensory, motor, and association cortex and in the hippocampus [14, 2, 16]. This has a stark impact on neural decoding: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is recalibration: periodically refitting the decoder on fresh, within-subject data [1, 4]. Recalibration requires ongoing labelled data from the individual, and it treats each subject in isolation.

We ask whether the code's cross-individual structure offers an alternative route to temporal robustness. Cortex-wide behavioral representations are known to be considerably shared among individuals [9, 6, 7], which allows models to decode behavior from unseen individuals [17]. The question we pose is temporal: does a decoder built from a population of other individuals resist drift better than a decoder built from the subject itself? If individual brains drift along idiosyncratic, partly independent trajectories, then averaging over many individuals should cancel the individual-specific component of drift. This prediction, to our knowledge, has not been examined.

We test it on BraiDyn-BC [5], a widefield calcium-imaging resource whose $\sim$15-day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time. Our contributions:

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

- **Pooling resists drift.** Across two weeks a population-pooled decoder loses less accuracy than a mouse's own (paired asymmetry $+0.017$ AUC, $p = 5 \times 10^{-5}$; Section 4.1). The effect varies in size across the four decoded targets, but those targets are chained within a trial and differ tenfold in event count, so we treat the pooled estimate as the result and the per-target ranking as uninterpreted (Section 4.2).
- **It is not a data-volume artifact.** A count-matched control isolates the cause. At a fixed number of training events, a decoder trained on one other mouse ages less than one trained on the subject's own, by $+0.007$ (lick), $+0.017$ (reward) and $+0.006$ AUC (tone). The gain therefore comes from training on individuals other than the subject, not from the larger training set pooling usually brings. Pooling over many mice rather than one lowers aging further on all four events, a smaller additional gain from a more diverse set of individuals (Sections 4.4–4.5).
- **Session-level validation is load-bearing.** Because trials within a recording session are not independent, a trial-level split inflates the personal decoder but not the pooled one, and so inflates the asymmetry from one side. We report the size of that inflation per event (Section 4.6); on lever pull it accounts for the entire apparent effect. We recommend session-held-out validation as the default for longitudinal decoding comparisons.
- **A drift-mitigation framing.** The early-week accuracy loss of cross-subject decoding is repaid as late-week stability, suggesting population priors as an alternative to per-subject recalibration. We release code and a no-raw-download streaming pipeline.

2 Related work

Representational drift and its mitigation. Drift is characterized within individual subjects [14, 2, 16, 13], and most approaches to mitigating it also operate within the subject. These include manifold stabilization [1], supervised and self-calibrating recalibration [4, 12], and adversarial or nonlinear alignment to the subject's own reference day [3, 11]. Each method uses later data from the same subject to stabilize the decoder. We instead ask whether data from other subjects can provide that stability.

Cross-subject decoding. Shared structure across individuals already supports cross-subject readout through manifold alignment [8], contrastive embeddings [15], and, on this cohort, atlas-aligned foundation models [17]. Multi-participant pooling can also reduce overfitting: HTNet, for example, pools participant data to avoid fitting a decoder to a small single-subject sample [10]. Our question differs in two ways. First, our count-matched control keeps the training-set size fixed; the relevant overfitting is therefore not a consequence of limited data, but of relying on subject-specific features that later drift. Second, prior work measures accuracy or transfer at a fixed time. We measure how that accuracy changes over days, asking whether a pooled decoder ages more slowly.

Conservation of cortex-wide dynamics. Cortex-wide activity contains shared, low-dimensional, movement-related components [9, 6, 7]. Their presence makes cross-subject pooling possible, although it does not alone imply that pooling will resist drift.

3 Data and methods

Dataset. BraiDyn-BC [5] (DANDI:001425, CC-BY) provides, for each mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ traces parcellated into 44 Allen Common Coordinate Framework regions at $\sim$30 Hz, together with time-aligned behavioral channels. The dataset includes 25 mice, each of which completed $\sim$15 daily operant sessions over $\sim$3 weeks. We stream the 44-region traces and behavioral channels without downloading the raw movies.

Features and events. We detect the onset of four events (lever pull, lick, reward, and tone) as rising edges, with a 1 s refractory period. For each onset we construct a 44-dimensional evoked feature by subtracting the mean pre-onset baseline (-0.5–0 s) from the mean post-onset $\Delta F/F$ response (0–1 s) in each parcel. Negative samples are drawn from matched-count windows more than 2 s from any onset *of the same event*; they are not screened against the other three events, and Section 4.2 reports how often they contain one. Every classification is a binary decode of one event against these matched negatives, scored by area under the ROC curve (AUC). The four decoders are therefore four independent binary detectors, not a four-way classifier: nothing in this paper asks a model to distinguish one event from another.

Decoders. The primary decoder is a standardized ℓ_2-regularized logistic regression on the 44-dimensional evoked feature. This is a linear readout of the parcel-space code: it recovers the component of behavior that is linearly present in the population average, and, because it has no capacity to memorize spatiotemporal detail, it cannot inflate an apparent generalization difference through overfitting. It is also the natural comparison to prior cross-subject decoders, which are overwhelmingly linear [8, 10]. Except where Section 4.7 varies it deliberately, every condition below uses this same readout, so the only variable is which data the decoder was trained on. For that section we additionally train a 1-D convolutional network over time and a GRU over the frame sequence on the full evoked window (-0.5 to $+1.0\,\mathrm{s}$, 45×44), each with a linear classification head, on identical events, blocks and splits.

Temporal blocks and aging. We group each mouse's sessions into an early block (days 1–5) and a late block (days 11–15) and report AUC under four conditions. The personal decoder is estimated by leaving out one whole early *session* at a time: for each early session s we train on the mouse's remaining early sessions and score the resulting decoder both on s (within-mouse same-block, WS) and on the late block (within-mouse across-block, WX), then average over s. Because WS and WX come from the same fitted decoders, they differ only in which data they are tested on. The pooled decoder is trained once on the other mice's early blocks and scored on the same held-out sessions s (LS) and on the late block (LX).

Holding out sessions rather than trials is essential and not a detail. Trials within one recording share illumination and hemodynamic state, slow arousal drift, and temporal autocorrelation, so a trial-level split lets the personal decoder train and test inside the same session. The pooled decoder is trained on other animals entirely and cannot benefit this way, so the inflation is one-sided and lands directly on the quantity of interest. Section 4.6 reports what that costs per event.

Aging is the accuracy lost when testing moves from the held-out mouse's early data to its late block while training stays fixed: personal aging is $\mathrm{WS} - \mathrm{WX}$ and pooled aging is $\mathrm{LS} - \mathrm{LX}$. The per-mouse aging asymmetry is $(\mathrm{WS} - \mathrm{WX}) - (\mathrm{LS} - \mathrm{LX})$; a positive value means the personal decoder ages more. We assess significance with a paired sign-flip permutation test across mice (20,000 permutations) and compute 95% confidence intervals with a cluster bootstrap over mice.

4 Results

4.1 A population-pooled decoder ages more slowly than a personal one

For each mouse, we keep the test data fixed (first the mouse's held-out early sessions, then its late block) and change only the source of the training data: either the mouse itself or the other 24 mice. The pooled decoder loses less accuracy across the two weeks (Table 1). Across all 100 mouse–target pairs the asymmetry is $+0.017$ AUC ($p = 5 \times 10^{-5}$), with the personal decoder aging more in 73% of pairs.

The effect is not uniform across the four targets: it is largest for reward ($+0.029$, $p = 0.0005$) and lick ($+0.020$, $p = 0.002$), marginal for tone ($+0.015$, $p = 0.08$), and absent for lever pull ($+0.003$, $p = 0.37$). We report this variation but deliberately do not build an interpretation on it. As Section 4.2 shows, the four targets are neither temporally separable nor matched in sample size, so the ranking across them cannot be read as a statement about which *behaviors* carry the effect. The pooled estimate is the claim we defend; the per-event column is data.

4.2 What the four targets are, and why we do not compare them

The four decoded events come from a single cued lever-pull trial structure, so they are chained in time rather than independent. Measuring onset times directly on eight sessions from eight mice: every reward onset has a lever-pull onset a median of $0.083\,\mathrm{s}$ away and a lick onset $0.117\,\mathrm{s}$ away, and 100% and 98% respectively fall inside the $[-0.5, +1.0]\,\mathrm{s}$ window from which a feature vector is built. Taken over all four targets, the fraction of positive windows containing at least one other event is 100% (reward), 84% (lick), 84% (tone) and 68% (lever pull). The reward decoder is therefore

Table 1: Personal vs. pooled aging across days (AUC lost from early to late test, holding training fixed), per event, $n = 25$ mice, with whole early sessions held out. Asymmetry = personal − pooled aging; a positive value means the mouse's own decoder ages more. p: paired sign-flip permutation. The rightmost column gives the same asymmetry under a trial-level split, the protocol this analysis replaces; the difference is the inflation quantified in Section 4.6.

Event	Personal	Pooled	Asymmetry (95% CI)	p	own ages more	trial-split
Lever-pull	-0.022	-0.025	$+0.003$ $[-0.004, 0.011]$	0.37	64%	$+0.014$
Lick	$+0.012$	-0.009	$+0.020$ $[0.008, 0.032]$	0.0023	80%	$+0.028$
Reward	$+0.017$	-0.012	$+0.029$ $[0.014, 0.046]$	0.0005	80%	$+0.032$
Tone	$+0.016$	$+0.001$	$+0.015$ $[0.001, 0.032]$	0.077	68%	$+0.022$
Pooled			$\mathbf{+0.017}$ $[0.010, 0.024]$	$\mathbf{5 \times 10^{-5}}$	**73%**	$+0.024$

never reading reward in isolation; its input window always contains licking and lever movement as well.

The negatives are also not event-free. Because they are screened only against the target's own onsets, they contain another event 33–57% of the time (56% for reward, 57% for tone). What these AUCs measure is thus closer to "this task moment versus another moment, which is often also a task moment" than to event versus rest, and the absolute values should be read accordingly.

The targets also differ in sample size by more than an order of magnitude: a median of 56 reward onsets per session against 107 (tone), 530 (lick) and 652 (lever pull). The per-event asymmetry runs roughly opposite to that count (Spearman $\rho \approx -0.8$ across the four targets, which with $n = 4$ is suggestive at most). A decoder fit to fewer events has more idiosyncrasy available to overfit, and therefore more for pooling to avoid, which would produce exactly this ordering without any of the behaviors differing.

We cannot separate these explanations with four targets, and we do not try. The consequence for the rest of the paper is deliberate: the pooled estimate over all 100 mouse–target pairs is what we defend, and the per-event breakdown in Table 1 is reported so the variation is visible, not as four independent findings. Where later sections note that an analysis selects reward and lick (Sections 4.3, 4.4, 4.7), that is a consistency check on the same targets, not evidence from independent behaviors.

4.3 The personal decoder's advantage erodes over the protocol

The block split compares two windows, but the effect should also appear as a continuous trend across all fifteen days. For each session day we compute the accuracy of the personal decoder and the pooled decoder and regress each on day; on early days the personal decoder is trained by leave-one-session-out, so it is never tested on a session it trained on. The personal decoder starts ahead: on day 1 it leads the pooled decoder by about 0.035 AUC, as expected for a decoder trained on the same animal. That lead then erodes. The gap narrows by 0.0014 AUC per day, pooled over all 100 mouse–target pairs ($p = 5 \times 10^{-5}$; the gap narrows in 65% of pairs), so by day 15 it has fallen to about 0.014 AUC: roughly 60% of the early-week advantage is gone. Averaged over the four events, the narrowing comes from the pooled decoder gaining accuracy over days (+0.0015 AUC/day) while the personal decoder stays flat (+0.0000 AUC/day); it is not that the personal decoder collapses. Per event, the personal decoder does lose accuracy for lick (−0.0010 AUC/day) and reward (−0.0016 AUC/day), and these are the two events where the narrowing is individually significant ($p = 0.007$ and $p = 0.001$). For lever pull both decoders improve with practice, and for tone the personal decoder is flat; in neither case is the gap slope significant (Fig. 1).

4.4 The robustness is not a data-volume artifact

A pooled decoder contains ∼24× more training events than a personal decoder. Its stability could therefore reflect data volume rather than pooling. We test this with a count-matched control (Fig. 2, left). For each held-out early session s of each mouse we set the training-set size to n, the number of events in that mouse's remaining early sessions, and compare three decoders: *self*, trained on those n events; *one*, trained on n events from a single other mouse; and *many*, trained on n events drawn

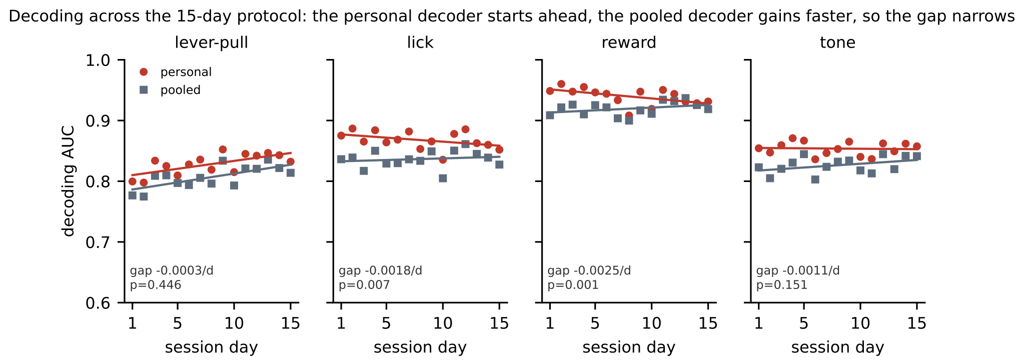

Figure 1: Decoding accuracy across the 15-day protocol for the personal decoder (own early data) and the pooled decoder (other mice's early data), with per-day points and linear fits. The personal decoder starts ahead, and the pooled decoder gains more over days, so the gap between them narrows. The narrowing is significant for lick and reward, the two events where the personal decoder also loses accuracy over days, and is not significant for lever pull or tone. The per-event gap slope and its p (paired sign-flip over mice) are shown.

across the other 24 mice. All three are scored on the same held-out session s and the same late block, so they differ only in whose data they were fit on.

For the events that showed an asymmetry in Section 4.1, the count-matched ordering reproduces it. The self decoder ages more than a decoder trained on one other mouse by $+0.007$ AUC for lick ($+0.012$ vs. $+0.005$), $+0.017$ for reward ($+0.017$ vs. $+0.000$), and $+0.006$ for tone ($+0.016$ vs. $+0.010$). Because both decoders see the same number of training events, data volume cannot explain these gaps: what distinguishes them is only whether the training data came from the subject. The reading we favour is that the mouse's own early decoder fits idiosyncratic features that later drift, features a decoder trained on another animal never had access to.

Lever pull again behaves differently: at matched count the self decoder ages *less* than the one-other decoder (-0.022 vs. -0.020), reversing the ordering. Consistently with Section 4.1, we do not claim the effect for this event.

The diversity ordering is the most robust result in this analysis: *many* ages less than *one* for all four events, including lever pull (-0.026 vs. -0.020; -0.003 vs. $+0.005$ for lick, -0.009 vs. $+0.000$ for reward, $+0.000$ vs. $+0.010$ for tone). Training on a diverse set of individuals rather than a single other individual therefore provides a smaller, additional benefit that does not depend on which event is decoded.

4.5 Diversity: a secondary scaling with the number of training individuals

We next vary the number of training mice, $N \in \{1, 2, 4, 8, 16, 24\}$ (Fig. 2, right). As N increases, pooled aging becomes slightly more favorable for lick (-0.001 to -0.010), reward (-0.004 to -0.010), and tone ($+0.003$ to -0.010). Lever-pull performance saturates with a single training mouse and remains near -0.025. The scaling effect is around 0.01 AUC, comparable to the 0.006–0.017 self-versus-one-other gap of Section 4.4 rather than small beside it. Most of the resistance to drift is already present when training on a single other individual, and additional individuals refine it further; this analysis never lets the decoder see the held-out mouse, so it is unaffected by the protocol issue of Section 4.6.

4.6 How much the validation protocol matters

Every number above holds out whole sessions. Repeating the identical analysis on the identical features with a trial-level split – shuffling trials pooled across the early sessions, so that a decoder may train on one part of a session and be tested on another – inflates the personal decoder's within-block accuracy by $+0.013$ (lever pull), $+0.010$ (lick), $+0.009$ (reward) and $+0.000$ AUC (tone). The

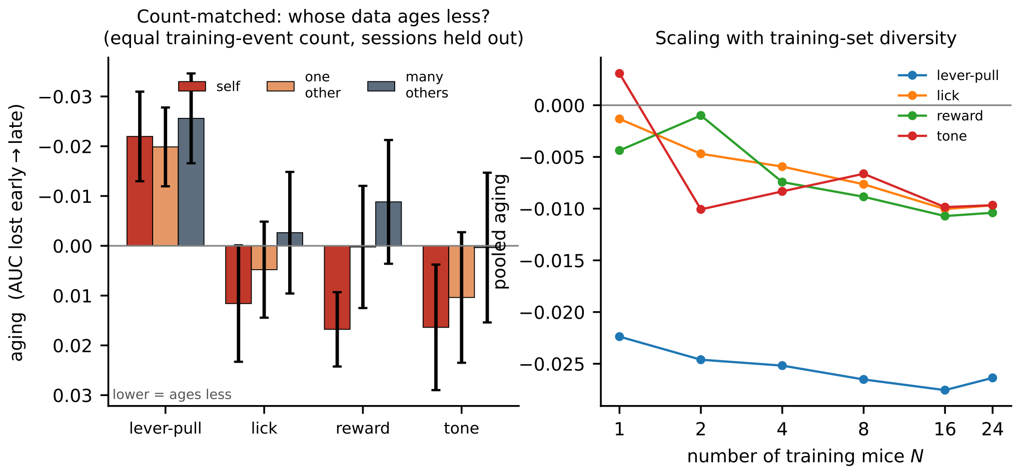

Figure 2: **Left:** count-matched control. With equal numbers of training events, decoder aging (AUC lost from early to late; lower values indicate greater robustness) is compared for the subject's own data (self), one other mouse, and many other mice. The self decoder ages, whereas the other-individual decoders do not. **Right:** pooled aging as a function of the number of training mice N. Greater diversity provides a small additional gain, with immediate saturation for lever pull.

pooled decoder is unaffected, because it is trained on other animals and has no session in common with the test data.

The consequence is that the inflation lands on one side of a difference between two arms. Asymmetries measured this way are $+0.014$, $+0.028$, $+0.032$ and $+0.022$ for lever pull, lick, reward and tone (rightmost column of Table 1), against $+0.003$, $+0.020$, $+0.029$ and $+0.015$ with sessions held out. For lever pull that difference is the entire effect: what looks like a significant $+0.014$ ($p = 0.001$) becomes $+0.003$ ($p = 0.37$). For reward, which has the largest true effect, it changes little.

The same reasoning applies to the count-matched control, where the natural implementation is to score the self decoder on the block it was fit on. That is not a cross-validated quantity at all, and it moves the self arm by -0.024 AUC on lever pull – enough to reverse the ordering the control exists to establish.

We report this because the pattern generalizes beyond our dataset. Any comparison of a within-subject decoder against a cross-subject one is exposed to it, since only the within-subject arm can share a session between train and test. Where trials are not independent – which for physiological recordings is the normal case – the split must respect the recording session, or the resulting asymmetry will be biased in a predictable direction.

4.7 The effect does not depend on the decoder

Every aging number so far comes from a linear readout, which could in principle be too weak to capture the drift-prone structure at issue. We therefore recompute the asymmetry with a 1-D convolutional network over time and a GRU over the frame sequence, on the full evoked window (-0.5 to $+1.0\,\mathrm{s}$, 45×44), under the same session-held-out protocol (Table 2). For the two events that carry the effect it survives every decoder: lick gives $+0.020$, $+0.012$ and $+0.010$ and reward $+0.029$, $+0.015$ and $+0.012$ under the linear, CNN and GRU readouts, all significant. Lever pull is null under all three ($p \geq 0.09$). The effect is therefore a property of the code rather than of the linear readout.

Two cautions. First, the corrected numbers do *not* show capacity amplifying the asymmetry: for lick and reward the nonlinear estimates are *smaller* than the linear one. Under a trial-level split the opposite appears to be true – the CNN reaches $+0.076$ on both lever pull and tone – but that is the leakage of Section 4.6, which a high-capacity model exploits more effectively than a linear

Table 2: Personal-minus-pooled aging asymmetry under three decoders, per event, $n = 25$ mice, all with whole early sessions held out. A positive value means the personal decoder ages more; p from a paired sign-flip permutation. The effect survives every decoder for lick and reward and is null under every decoder for lever pull. The nonlinear estimates are smaller than the linear ones, not larger: the apparent amplification with capacity reported under a trial-level split is the leakage of Section 4.6.

Event	Linear	1-D CNN	GRU
Lever-pull	$+0.006$ (0.11)	$+0.021$ (0.09)	$+0.016$ (0.16)
Lick	$+0.020$ (0.004)	$+0.012$ (0.007)	$+0.010$ (0.024)
Reward	$+0.029$ (0.0007)	$+0.015$ (0.020)	$+0.012$ (0.006)
Tone	$+0.016$ (0.09)	$+0.047$ ($<10^{-4}$)	$+0.020$ (0.014)

one. Holding out sessions removes 43–73% of the CNN's asymmetry against 30–48% of the linear model's on the same events. A capacity comparison run on a trial-level split therefore measures capacity to exploit within-session structure at least as much as capacity to overfit drift.

Second, tone does not resolve cleanly: it is marginal under the linear readout ($p = 0.09$) but clearly significant under the CNN ($+0.047$, $p < 10^{-4}$) and GRU ($+0.020$, $p = 0.014$). We leave it as unresolved rather than counting it for or against.

5 Discussion

A subject's own decoder overfits idiosyncratic features of its early sessions, some of which later drift. A decoder that has never seen the subject must instead rely on components of the code that are shared across individuals and more stable over time. The count-matched control supports this reading directly: at equal data, other-individual data ages less than the subject's own for the events where we detect the effect. Population structure is therefore not only a resource for cross-subject transfer, but also for temporal robustness. The early-week accuracy cost of cross-subject decoding is recovered over time as the pooled decoder remains stable. Unlike within-subject recalibration, this approach does not require newly labelled data from the individual at test time.

The two events that carry the effect, lick and reward, are also the two whose personal decoders lose accuracy over the protocol; the two that do not, lever pull and tone, are the two where the animal's performance is still improving and both decoders gain. This is consistent with the account above – there is more subject-specific drift to avoid overfitting where drift is actually occurring – but we did not predict the split in advance and treat it as an observation rather than a tested hypothesis.

Limitations. The clearest limitation is what our four targets are. They are not four independent behaviors but four thresholded channels of one cued lever-pull trial, overlapping in time to the point where every reward window also contains licking, and differing tenfold in event count (Section 4.2). We therefore cannot say which behavior the effect belongs to, nor exclude that the per-target ranking simply tracks how much data each decoder had to overfit. A design with temporally separated, count-matched events would settle both, and is the experiment we would run next. The pooled estimate over all 100 mouse–target pairs should be read as an average over four correlated readouts of one task, not as a uniform property of cortex. We measure at the mesoscale, using parcel-averaged activity, so whether the same result holds for single-neuron codes – where representational drift is best characterized – remains unknown. The analysis covers a two-week interval, discrete task events, and one species; longer horizons, graded behaviors, other species, and other recording modalities should be tested. Our conclusions depend on within-versus-pooled and early-versus-late comparisons rather than on absolute AUC. Although the count-matched control rules out training-set size and is consistent with self-overfitting, it does not identify which components of drift are individual-specific. Finally, tone remains unresolved: marginal under a linear readout and significant under both nonlinear ones, it is the one event where the choice of decoder changes the answer, and we do not count it either way.

Reproducibility

BraiDyn-BC is publicly available through DANDI:001425. We release the analysis code and a streaming feature pipeline that reproduces all reported results from the parcellated traces without downloading the raw imaging data. Section 3 and the released code specify all data splits, permutation seeds, and hyperparameters.

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